

SENTINEL

Integrated Video Management and Analytics Platform

A hybrid of Model 1 (Centralised CCTV Registry and GIS Mapping) and Model 2 (Unified Viewing and Metadata Analytics), with event-triggered evidence capture (Pipeline 3) designed and validated separately. Model 3 federation is the documented integration path.

Police

GSRTC

Municipal

Panchayat

Health

One platform. Five departments. Watching is not storing.

Gujarat Police Innovation Challenge 2026 · Home Department / SCRB · Submission by Adi

The problem, and what is actually graded

TODAY

- 26 government departments run independent, standalone CCTV systems
- Mixed analog and IP cameras, different vendors and VMS platforms, incompatible feed protocols
- Storage fragmented: cloud for some, local for others, retention from 7 days to over 15
- Sites as much as 1,000 km apart, from the border districts to Somnath and Dwarka
- Tracing one vehicle means asking each department separately, by hand

THE LIVE TEST CASE

- **A registration number is handed over on evaluation day.**
- Identify, trace and present that vehicle's movement across the integrated network
- Complete route: timestamped, location-wise, across camera locations and departments
- Continuous cross-referencing of live feeds against a watchlist, with automated real-time alerts
- A working system. Mock-ups and concept videos are not considered

Proposed model: a hybrid, with justification

MODEL 1 · mandatory

- Camera registry, the control plane
- GIS console: department layers, coverage sectors, activity, gaps
- Onboarding by form, CSV and API
- Built and demonstrated

MODEL 2 · viewing + analytics

- Direct pull, departments untouched
- Two different systems in one viewer
- ANPR metadata, no central video
- Built and demonstrated

PIPELINE 3 · evidence

- Rolling buffer, promote on a match only
- 60 s clip, SHA-256, audit row
- Designed and validated separately
- Not built in the demo

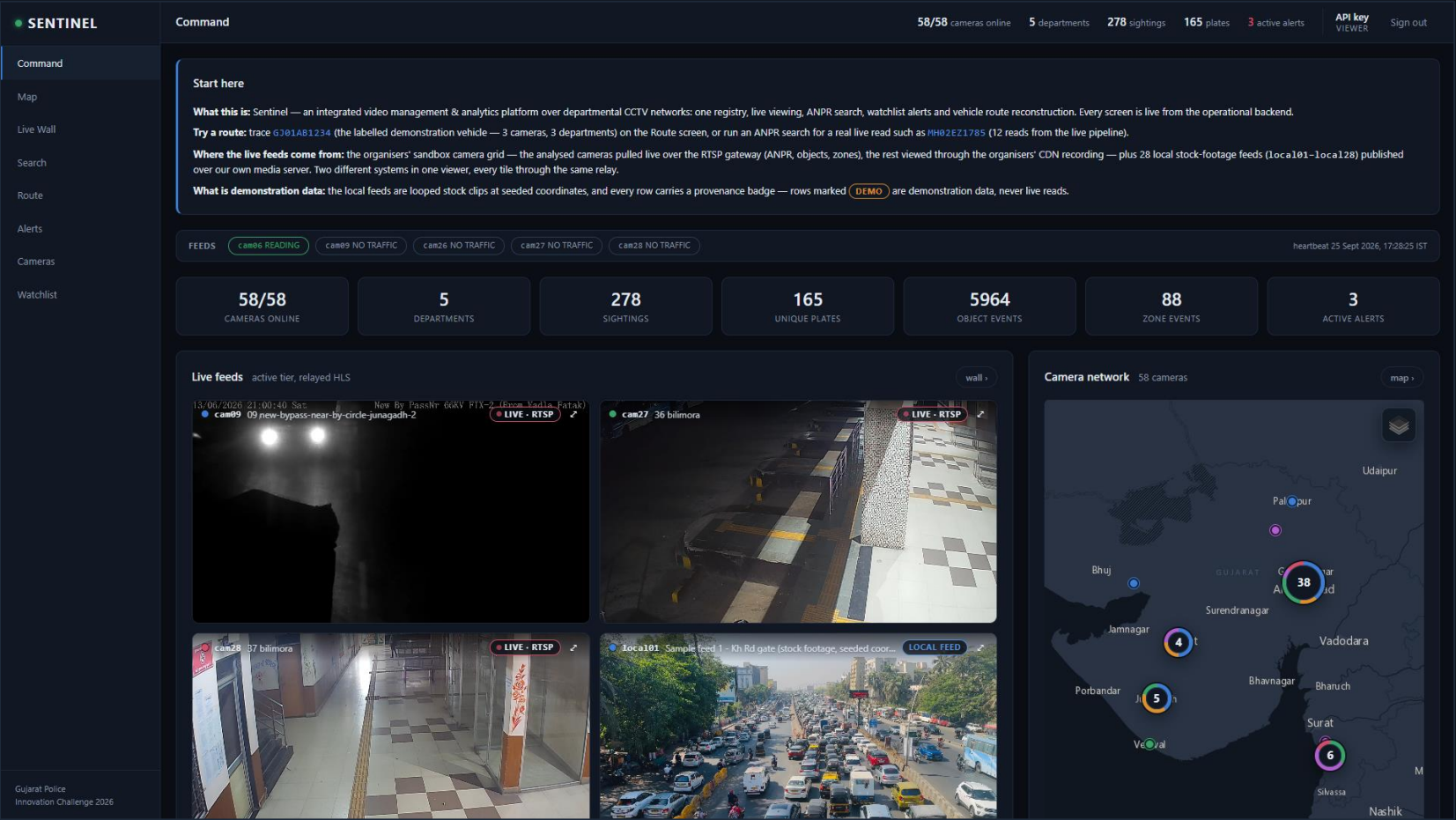
MODEL 3 · federation

- Adapter per vendor VMS
- Department-side collector for NAT and SDK-only sites
- Answers heterogeneity, not scale
- Integration path in the HLD

Why not Model 4, Central VMS and AI Platform: 80,000 cameras at 3 Mbps is about 240 Gbps sustained into one facility, a procurement and physical-plant problem before it is a budget line. Analysing at the edge sends only text rows and plate crops, about 23 Mbps, roughly 10,000 times less. [model]

What we built: the Command view

One screen: live tiles, GIS, alerts, plate reads, analytics and feed health, all served by the operational backend



Command view, live: sandbox cameras cam09, cam27 and cam28 over RTSP and sample feed local01, all through the relay (captured 25 Sep 2026)

- 1 **Live tiles** Analysed feeds relayed, not recorded: a self-overwriting 20 s window is the only video on disk.
- 2 **GIS** 58 cameras: the organisers' 30 and 28 local sample feeds, clustered by city.
- 3 **Alerts** Alert rows tailed from the database and pushed over Server-Sent Events in about 2 s [estimate].
- 4 **Feed status** Reading, no traffic or feed down, judged by each pull's own freshness.
- 5 **Start here** The first screen says which data is demonstration data. Demo rows carry a DEMO badge.
- 6 **Reads and analytics** Latest plate reads, object, person and zone-event counts from the live detector.

Architecture: one pull, three pipelines, one registry

MODEL 1 REGISTRY + GIS the control plane: id, department, location, bearing, field of view, stream URLs (never credentials), codec, health, tier, zones. Nothing downstream hard-codes a camera.

CAMERA GRID > ONE pull per camera (RTSP over TCP) > NODE

PIPELINE 1 · LIVE VIEW

- Relay to the control room: Command and the Live Wall
- hls.js tiles; the API holds the upstream session
- Persists: a self-overwriting 20 s relay window only

relayed, not recorded · built

PIPELINE 2 · AI ANALYTICS

- Motion gate, detect, track, crop, OCR
- Match against the LOCAL cached watchlist
- Persists: text rows plus a plate crop of about 2 KB

text and crop only · built

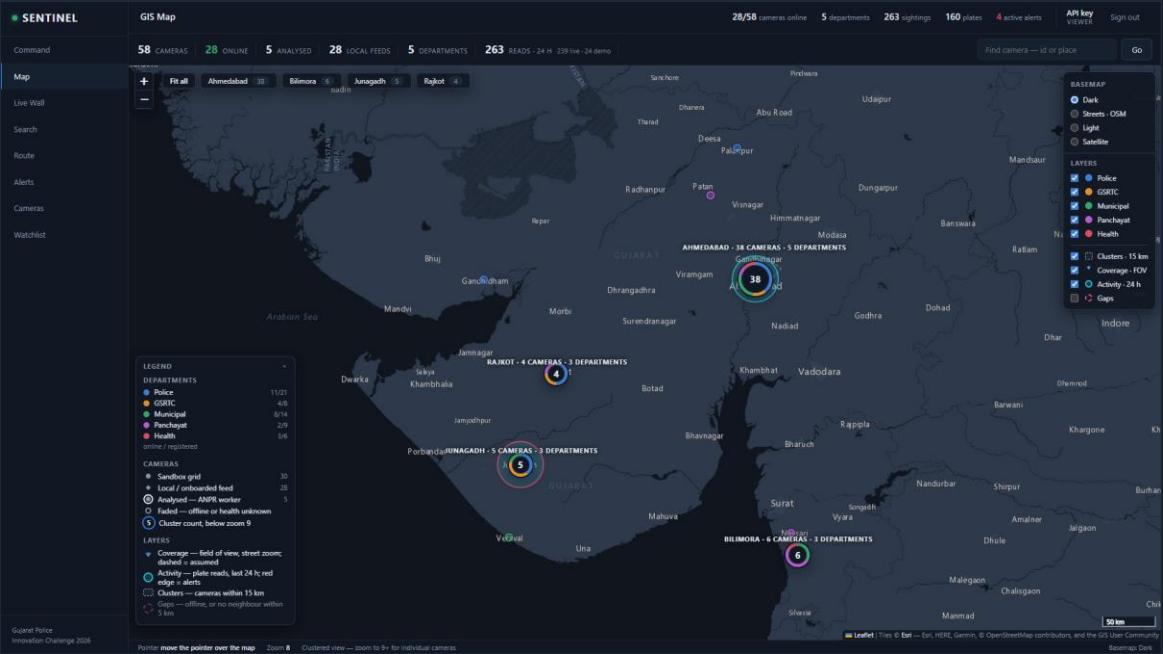
PIPELINE 3 · EVIDENCE

- Fixed-size rolling buffer per camera
- Promote 30 s either side, on a watchlist match only
- Persists: clip, SHA-256, audit row

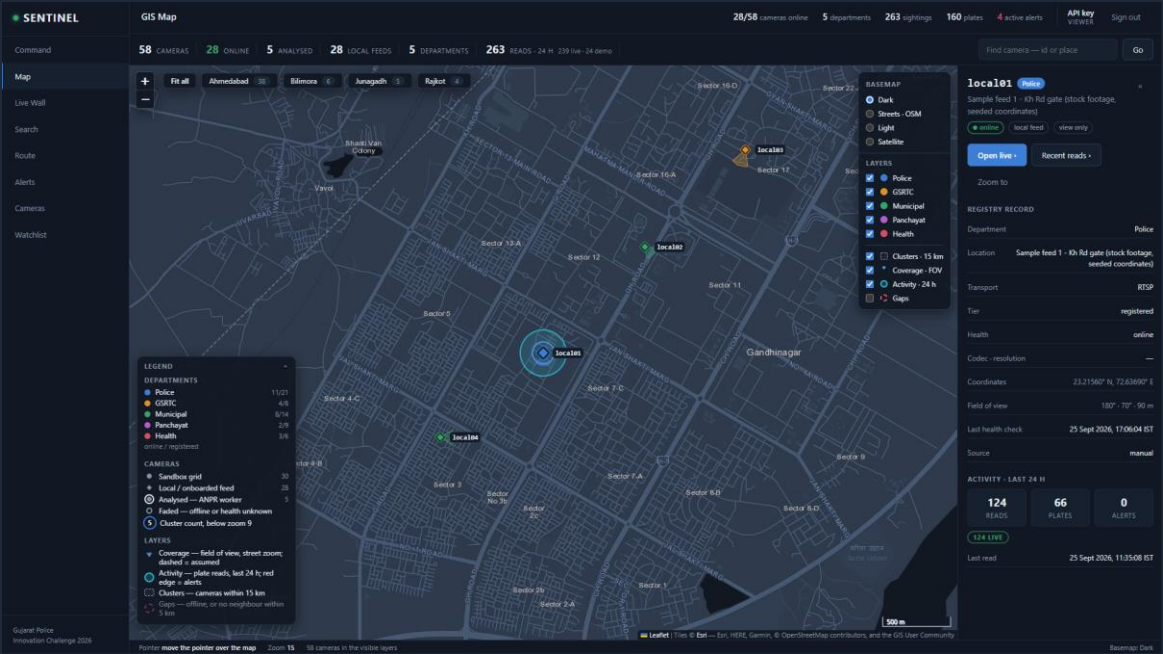
designed and validated separately; not built in the demo

Governing rule: video becomes permanent only where a specific, logged, auditable watchlist match justifies it. In the demonstrated system, where Pipeline 3 is not built, no video is kept beyond the 20-second relay window.

Model 1: registry, GIS console and the named deliverables



GIS console, live: all 58 cameras clustered by city, department layers, basemap choice and legend

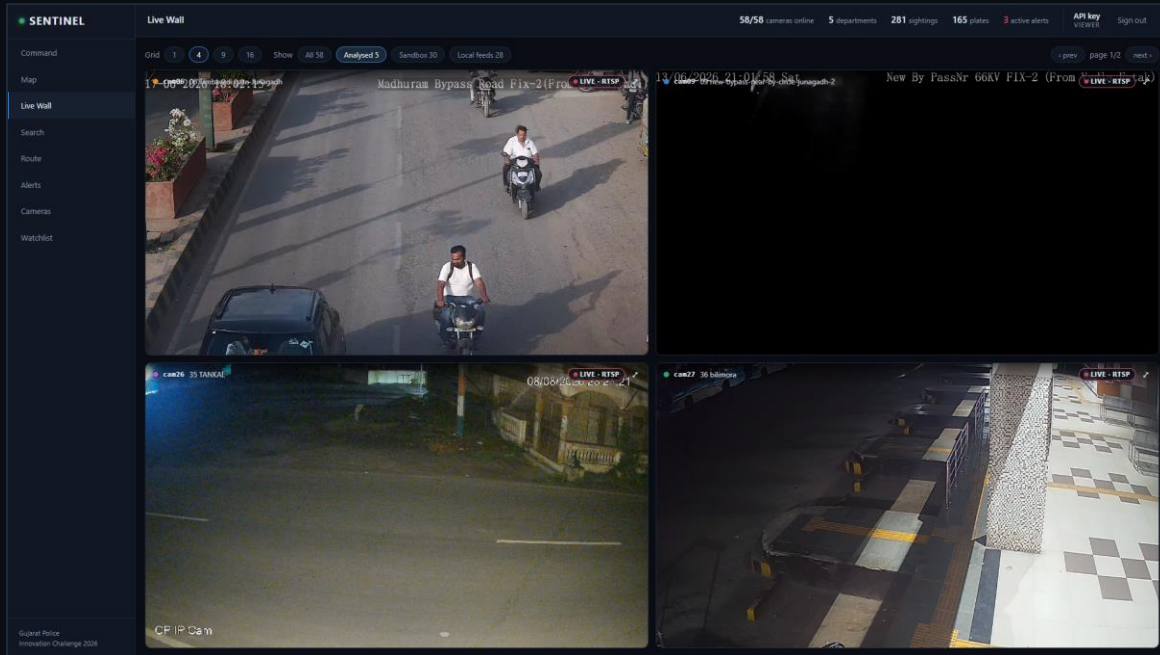


Zoomed to Gandhinagar: pins by department, field-of-view sectors, 24 h read activity and the camera's registry record

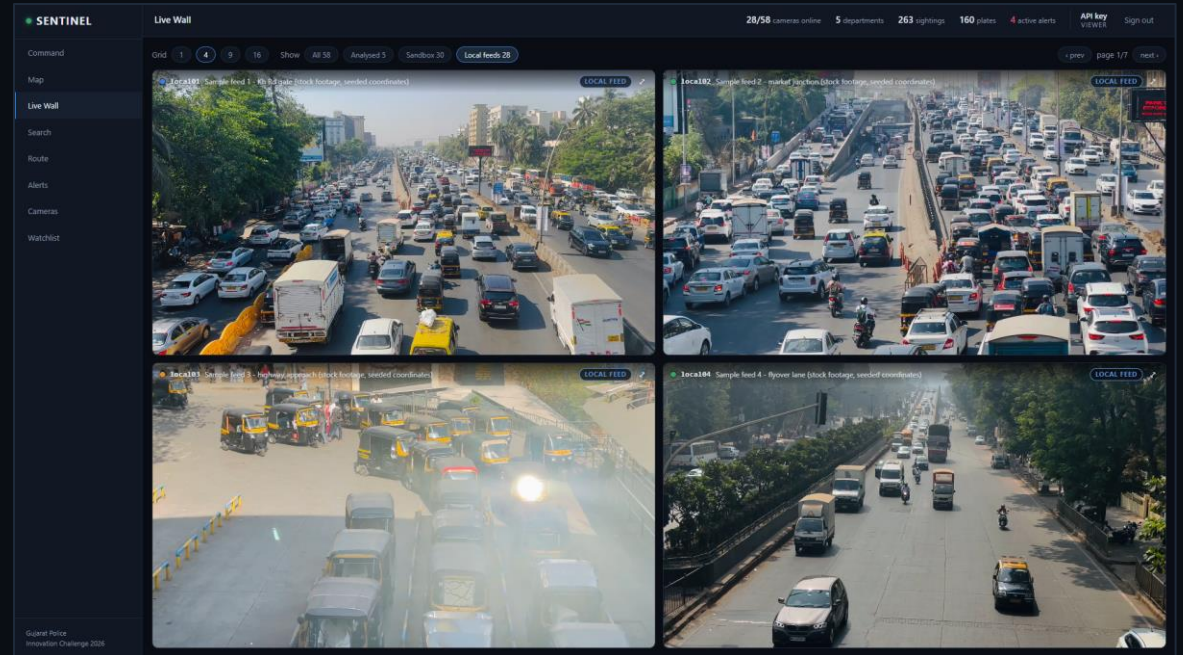
MODEL 1 DELIVERABLES: BUILT

- Registry portal with a GIS console: street, light and satellite basemaps; department layers, coverage sectors, activity, cluster hulls, coverage gaps
- Onboarding by form, bulk CSV (each row accepted or rejected with a reason) and POST /api/cameras, every action audited
- 58 cameras registered: the organisers' 30 across 5 departments and 28 local sample feeds; registry API documented as OpenAPI; gap-analysis report; health monitoring
- Disclosed: the catalogue carries only id and name. Departments and coordinates of the sandbox cameras and of the sample feeds are seeded in committed files.

Model 2: two different systems in one viewer



System A, the organisers' sandbox: four analysed cameras pulled live over RTSP and played through the relay



System B, a local camera server: 28 stock-footage feeds (sample clips at seeded coordinates, labelled on every tile)

- One pull per camera, RTSP over TCP and never UDP; decoder warnings on join are logged, never fatal
- All timing from PTS; jittered exponential backoff (2 s base, 30 s cap) and a stall watchdog on every pull
- Consume-only: no publishing, no control-API calls, no footage download
- Every registered camera opens on the Live Wall through the relay: the node's own pull, the organisers' CDN recording, or the local server's HLS
- Only the tiles on screen hold a stream; the relay sits behind the login and is rate-limited
- The sample feeds are stock footage, not sandbox footage and not filmed by the team

Video analytics: ANPR, objects, intrusion, tracking

The screenshot displays the SENTINEL ANPR Search interface. The top navigation bar includes 'SENTINEL', 'ANPR Search', and various system metrics (28/58 cameras online, 5 departments, 263 lightings, 160 plates, 4 active alerts). The left sidebar contains navigation links: Command, Map, Live Wall, Search, Route, Alerts, Cameras, and Watchlist. The main content area is titled 'ANPR search' and includes a search bar with the registration number 'GJ11S7924'. Below the search bar, there are filters for 'FULL REGISTRATION' and 'ANPR-tolerant'. The 'REFINE' section allows filtering by 'CAMERA', 'MIN CONFIDENCE', 'PROVENANCE', and 'VEHICLE CLASS'. The 'Reads' section shows 10 reads for the registration number, with a table of results. The table columns are: PLATE, MATCH, CONFIDENCE, CLASS, CAMERA - DEPARTMENT, SEEN (IST), and DATA. The table lists 10 reads for the registration number GJ11S7924, showing matches, confidence scores, vehicle classes, and camera details.

PLATE	MATCH	CONFIDENCE	CLASS	CAMERA - DEPARTMENT	SEEN (IST)	DATA
GJ11S7924	match	0.99	motorcycle	cam06 - GSRTC	25 Sept 2026, 16:54:33 IST	LIVE route
GJ11S7924	match	0.91	motorcycle	cam06 - GSRTC	25 Sept 2026, 16:50:50 IST	LIVE route
GJ11S7924	match	0.87	truck	cam06 - GSRTC	25 Sept 2026, 14:48:10 IST	LIVE route
GJ11S7924	match	0.84	truck	cam06 - GSRTC	25 Sept 2026, 14:44:26 IST	LIVE route
GJ11S7924	match	0.88	motorcycle	cam06 - GSRTC	25 Sept 2026, 14:10:11 IST	LIVE route
GJ11S7924	match	0.96	truck	cam06 - GSRTC	25 Sept 2026, 14:07:30 IST	LIVE route
GJ11S7924	match	0.94	bus	cam06 - GSRTC	25 Sept 2026, 13:41:45 IST	LIVE route

ANPR on the live sandbox: every read of GJ11S7924 on cam06, the OCR text kept beside the coerced plate, with crop, confidence, class and provenance

Facial recognition: described, not built. Face detection at the edge, then embedding, then matching against an authorised gallery only, with the same edge-cached posture. No general-population face database. Non-match embeddings are not retained. Every match is logged.

- 1 Motion gate** MOG2 skips static frames; measured skip rate 0 to 54% by camera [measured]
- 2 Detect** YOLOX-S (Apache-2.0) on ONNX Runtime, on the GPU through DirectML. Never Ultralytics (AGPL)
- 3 Track** A greedy IoU and centre-distance tracker on PTS deltas: an IoU tracker, not ByteTrack
- 4 Crop and OCR** The vehicle crop, never the frame, upscaled and read by PaddleOCR PP-OCRv5 mobile
- 5 Plate grammar** Indian format with position-aware coercion (GJ11S7924 stored as GJ11S7924); a track's reads are voted; raw OCR kept
- 6 Events** Object and person counts, intrusion polygons, directional line crossing

MEASURED ON THE LIVE SANDBOX, 25 SEP (10-minute run, five cameras over RTSP, laptop GTX 1650): inference 0.6 to 1.5 fps per camera; peak GPU memory 119 MiB; peak RAM 1,749 MB; plate-read rate 0.062 [measured]. The ANPR viability gate passed; the binding stage was CPU OCR, not the GPU.

Watchlist correlation and real-time alerting

The screenshot shows the 'Live Alerts' section of the SENTINEL interface. It displays a stream of alerts for the vehicle GJ01AB1234. Each alert row includes a plate crop, severity (HIGH), category (stolen_vehicle), match type (ambiguity d=0), a DEMO badge, and a route link. The alerts are timestamped and include location details.

Plate	Severity	Category	Match Type	Alert Type	Location	Timestamp
GJ 01 AB 1234	HIGH	stolen_vehicle	ambiguity d=0	DEMO	cam09: 09 new-bypass-near-by-circle-junagadh-2	25 Sept 2026, 16:43:49 IST
GJ 01 AB 1234	HIGH	stolen_vehicle	exact	DEMO	cam09: 09 new-bypass-near-by-circle-junagadh-2	25 Sept 2026, 16:37:49 IST
GJ 01 AB 1234	HIGH	stolen_vehicle	exact	DEMO	cam10: 10 char-chowk-road-2-junagadh Municipal	25 Sept 2026, 16:29:49 IST
GJ 01 AB 1234	HIGH	stolen_vehicle	exact	DEMO	cam06: 06 Timbarwad gate-junagadh GDRIC	25 Sept 2026, 16:22:49 IST

Alert stream: plate crop, severity, category, match type and a DEMO badge on every demonstration row

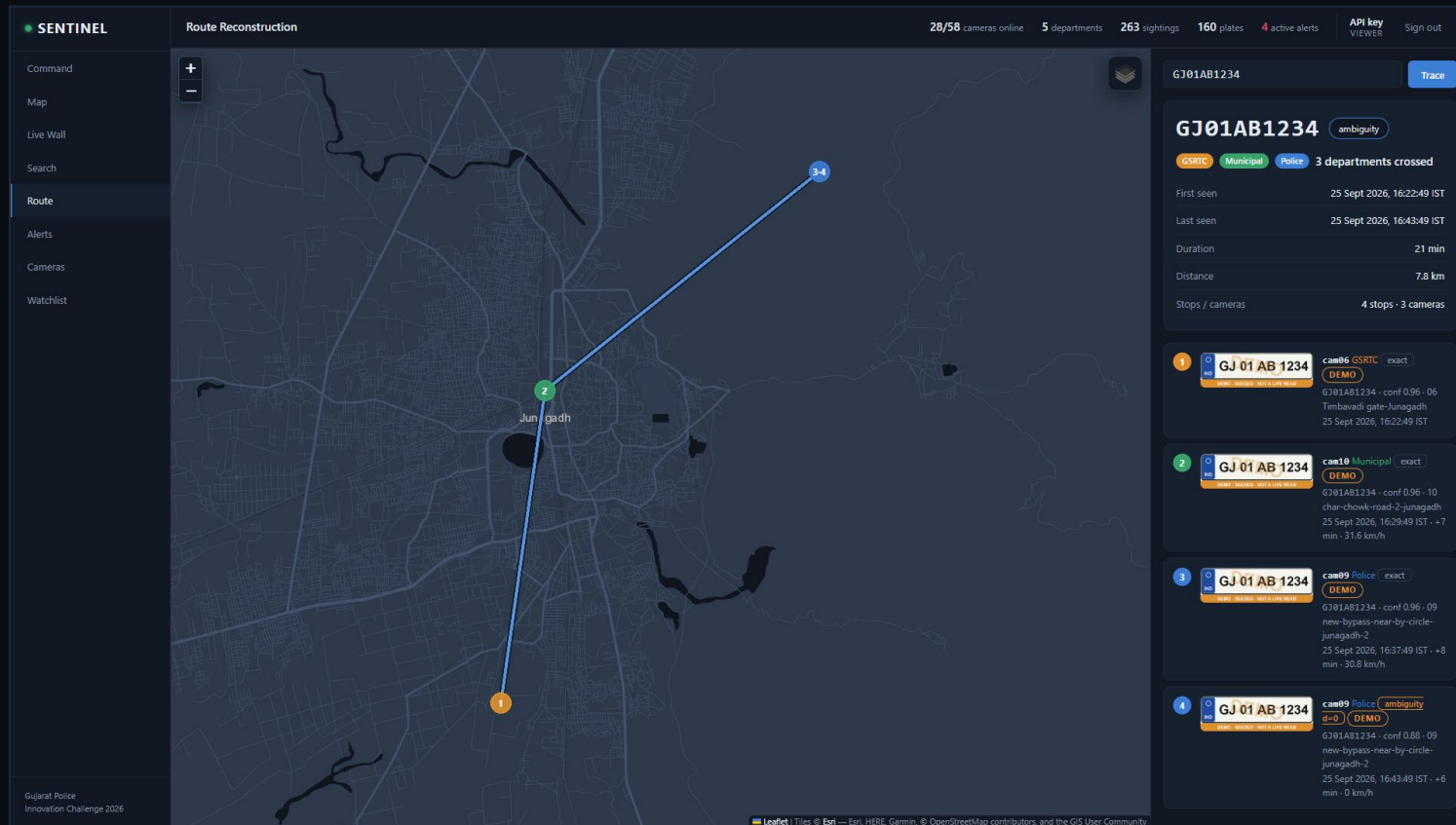
The screenshot shows the 'ANPR Search' section of the SENTINEL interface. It displays search results for the query GJ01A81234. The results show that the system corrected the mistyped query to GJ01AB1234 and found matches in the watchlist. The results table includes plate, match, confidence, class, camera, department, seen (IST), and data.

Plate	Match	Confidence	Class	Camera	Department	Seen (IST)	Data
GJ 01 AB 1234	exact	0.88	car	cam09	Police	25 Sept 2026, 16:43:49 IST	DEMO route
GJ 01 AB 1234	ambiguity d=0	0.96	car	cam09	Police	25 Sept 2026, 16:37:49 IST	DEMO route
GJ 01 AB 1234	ambiguity d=0	0.96	car	cam10	Municipal	25 Sept 2026, 16:29:49 IST	DEMO route
GJ 01 AB 1234	ambiguity d=0	0.96	car	cam06	GDRIC	25 Sept 2026, 16:22:49 IST	DEMO route

A mistyped query (GJ01A81234) still finds the watchlisted vehicle: the 8/B look-alike is corrected by position

- **Matching is local, against a cached watchlist: never a per-detection call to VAHAN or eGujCop**
- Three match types in order: exact, OCR-ambiguity (O/0, I/1, S/5, B/8, Z/2, G/6, Q/0), then confusion-weighted fuzzy
- Alerts fire on exact and ambiguity; fuzzy is shown and flagged, not alerted by default; partial reads never alert
- Sighting written before matching; alert row written before broadcast; the API tails the alerts table and pushes Server-Sent Events
- Five-minute cooldown per plate per camera, derived from the database rather than process memory
- Acknowledgements persisted and audited under the operator's name; every plate lookup is audited too

Route reconstruction: the scored capability

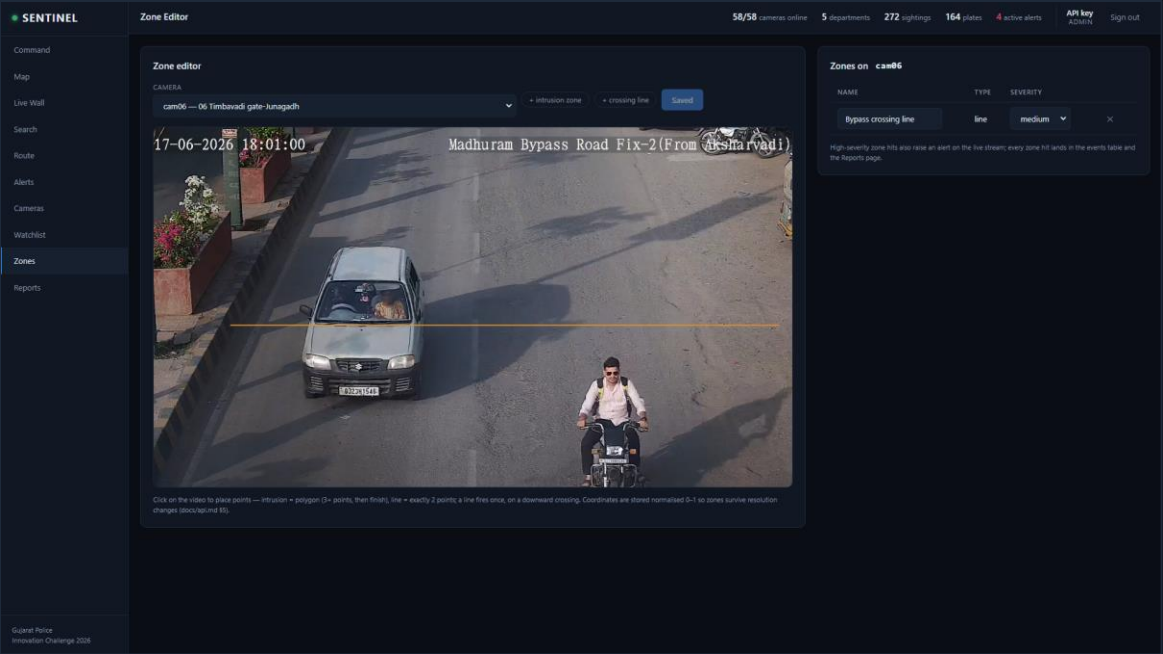


GJ01AB1234, the labelled demonstration vehicle: 4 stops, 3 cameras, 3 departments, numbered in time order with crops and implied speeds

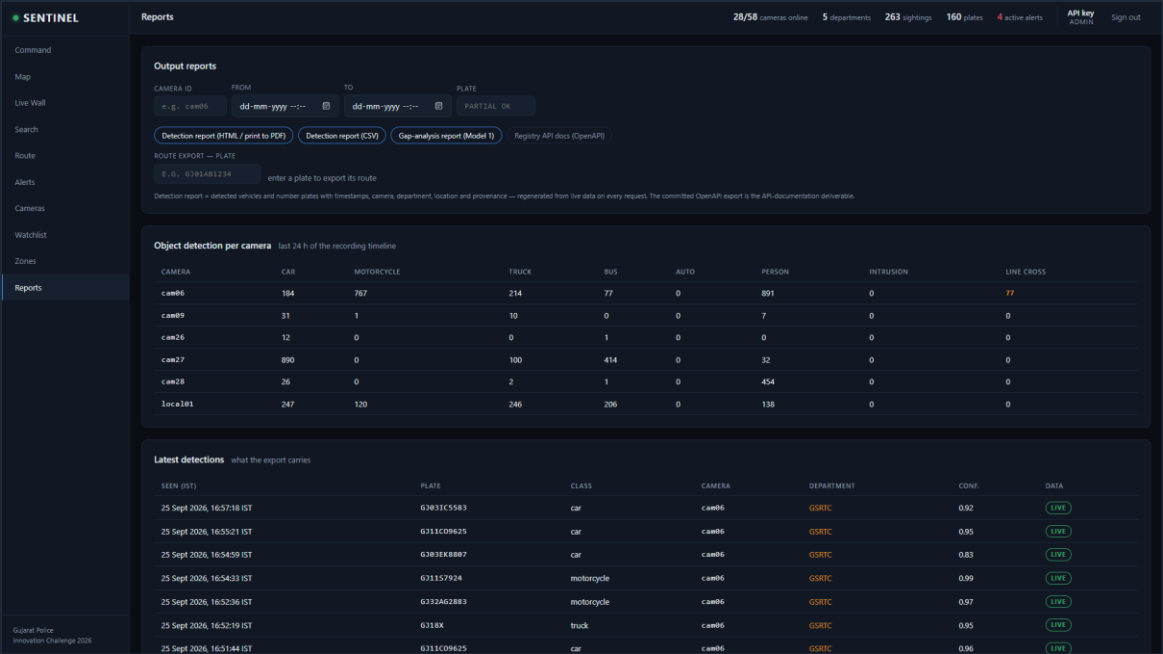
GET /api/plates/{plate}/route

- Ordered stops: camera, department, coordinates, timestamp, crop, match type and provenance
- Elapsed time and implied speed only within one clock; an implausible speed flags the stop as suspect
- Fuzzy candidates are surfaced and marked, never silently merged
- **departments_crossed is the integration proof**
- Coverage gaps are shown rather than hidden
- Route export and detection report as CSV and printable HTML, each with a provenance column
- Disclosed: no real vehicle has been read on two sandbox cameras. The route shown is demonstration data, labelled DEMO on every screen and export.

Beyond ANPR: intrusion zones, object counts, output reports



Zone editor on the live cam06 picture: the crossing line that fired 53 line-crossing events on the live sandbox during the measured afternoon run



Reports: detection report as CSV and printable HTML, gap analysis, OpenAPI; object counts per camera; every row labelled live or demo

- Intrusion: a tracked object's foot point must be inside a zone for two consecutive sampled frames before it fires; a line crossing is measured against the last confirmed side, with direction. High-severity zones raise alerts on the same stream.
- Object and person detection: per-class events, throttled to one per class per camera every 5 s of stream time, counted on Command and in Reports. Zones are stored in normalised 0 to 1 coordinates, so they survive a change of resolution.

Why edge-first: what crosses the WAN at 80,000 cameras

All figures are [model], from one plate read per camera per minute, to be replaced by measurement in the pilot

240 Gbps

raw video, centralised (1080p at 3 Mbps)

>

23 Mbps

text rows and plate crops, edge-first

10,000x

less WAN load

GPU FLEET: SIZING FOLLOWS ENGINEERING CHOICES [model]

Naive: every frame, full-frame ANPR	30 / GPU	2,667 GPUs
5 fps sampling	50	1,600
plus motion gating	100	800
plus cascade and INT8	150	533
plus ROI and tuning	200	400

DECODE IS USUALLY THE BINDING CEILING

- Every stream is H.264 or H.265 decoded before any model sees it, and decode engines cap at roughly 20 to 40 sessions per GPU [estimate]
- At 30 per GPU, decode alone needs 2,667 GPUs, so tuning inference to 200 streams per GPU leaves the tensor cores about 85% idle [model]
- Mitigation: decode only the frames that are inferred, prefer H.265, CPU-decode low-fps tiers, provision against the ceiling that binds
- The laptop run adds a lesson: its binding stage was OCR on the CPU, so a node budgets OCR on the accelerator too

Storage at scale

Per day, statewide. [model] from one plate read per camera per minute

3.46 TB

detection snapshots as full frames

0.23 TB

plate crops of about 2 KB instead, 15 times less

220 GB

evidence clips at 10,000 hits per day

- Retention tiers: hot for 7 days, warm for 90, then a cold or frozen archive. Sighting records run to about 115 million a day, 42 billion a year and 21 TB a year indexed [model], so shard and tier from day one. Retrofitting that after the index is built is a migration with no good window.
- Full frames only for confirmed watchlist hits, where the evidence clip exists anyway. Unmatched sighting crops expire in days rather than years.
- Evidence capture (Pipeline 3) is designed and validated separately and is not built in the demo: a fixed-size rolling buffer per camera stays flat and promotes a hashed clip only on a match. In the demo, the only video on disk is the self-overwriting 20-second relay window per analysed camera.
- Retention by data class, purged on a schedule: proposed non-hit crops 30 days, sighting rows 180 days, hit evidence for the life of the case [estimate]. Each department's own 7 to 15-day video retention is untouched. The purge is part of the pilot build.

Cybersecurity, privacy and auditability

Credentials

Environment or secret store only. Never hard-coded, logged, persisted or shown unmasked; the registry holds URL templates with placeholders filled in memory. A vault with per-department rotation in production.

Login and roles (built)

People sign in as viewer, evaluator or admin. Passwords only as script hashes; server-side sessions in HttpOnly, SameSite=Strict cookies expiring after 8 h; lockout after five failures. Department scoping is added in production.

Encryption

TLS on every hop the platform controls. When the demo is published, it goes out over an outbound-only tunnel with TLS terminated on the laptop. Encryption at rest for the evidence store and the databases in production.

Hardening (built)

Nothing is reachable without signing in except the login page, /api/health and static assets. Content-Security-Policy, nosniff, frame denial, rate limits on expensive reads. Only the API port is published.

Audit (built)

An append-only audit table names the user for every onboarding, watchlist change, acknowledgement, zone edit and login, and for every plate lookup, search and route export. Pipeline 3 adds hashed evidence rows.

Privacy

Relayed, not recorded: text and a 2 KB crop per read. Retention by data class, designed to the DPDP Act, 2023 duties. Facial recognition only ever against an authorised gallery.

Deployment, scalability and the phased rollout

EDGE

district or department

pull, decode, motion gate, detect, OCR, local watchlist match, ring buffer. Emits text and crops

REGIONAL

district cluster

aggregation, regional search index, evidence store, model rollout, camera health scoring

CENTRAL

SCRB or state data centre

statewide search, cross-region route correlation, watchlist master, and VAHAN, SARTHI, eGujCop, AFIS and NAFIS enrichment

RESILIENCE AND DAY-2 OPERATIONS

- Reconnect storms: jittered backoff, admission control, staggered cold start, and a circuit breaker per department
- Split-brain ownership: distributed camera leases, so never two pulls and never duplicate alerts
- The alert path is the real single point of failure: detections are persisted before alerting, with synthetic end-to-end tests
- A quality score per camera including drift detection, feeding a ranked maintenance worklist. Shadow then canary model rollout. GitOps configuration

PHASED STATEWIDE ROLLOUT

- 0 **Sandbox, now** 30 sandbox cameras and 28 sample feeds, one node
- 1 **Pilot district** one department's real cameras, measured streams per accelerator, Pipeline 3 built
- 2 **Multi-district** 3 to 5 departments, federation adapters and collector
- 3 **Statewide** towards 80,000 cameras, HA and DR, enrichment integrations

What this system does not do, and why

No evidence clips in the demonstrated system

Pipeline 3 is designed and validated separately and is not built. No alert links a clip, and no video is kept beyond the 20-second relay window.

No arbitrary rewind, no central recording of all video

Both the privacy posture and the 240 Gbps arithmetic rule it out. Video is captured on justified cause only.

No real cross-camera route on the sandbox feeds

No real vehicle has been read on two sandbox cameras. The route shown is a demonstration vehicle labelled DEMO everywhere; the scored test uses the organisers' own vehicle.

No Model 3 federation, no live facial recognition

The sandbox holds no departmental VMS to federate, so Model 3 is the documented integration path. Facial recognition is described with its privacy controls, not built and not claimed.

Plate recall on these feeds is low, and stated

Wide overview cameras put plates at 10 to 25 px [estimate]. Plate-read rate 0.062, 12 reads in 195 vehicle tracks [measured]; cam06 gave 55 real reads over the afternoon, and the ANPR viability gate passed on that evidence.

Demonstration data, disclosed

Sandbox geography is seeded (the catalogue has id and name only); the 28 sample feeds are stock footage at seeded coordinates; the demo vehicle is labelled. Figures are marked [measured], [model] or [estimate]; the 30 versus 50 camera gap is recorded.

Operational benefits and impact on policing

Hours become seconds

A registration number returns a timestamped, location-wise route across departments on one screen, instead of a manual request to each of them.

One platform, 26 departments

Departments keep their systems untouched. The connector ladder and the department-side collector reach vendor-locked and NAT-bound sites.

Alerts operators act on

Local matching, cooldowns, a plate crop on every alert and an acknowledgement trail. No alert floods, and no neighbour's car alerted on a fuzzy read.

Evidence that holds up

Built: every lookup, change and acknowledgement is audited under a named user. Designed for the pilot: clips promoted on justified cause, each with a SHA-256 and an audit row.

Coverage that stays usable

Health scoring, drift detection and a ranked maintenance worklist for the roughly 2% of cameras that are always broken [model].

A cost position the State can own

Linked components are Apache, MIT and BSD; ffmpeg runs as a separate GPL process. No per-camera licence avoids about 4 crore rupees a year at 500 rupees per camera [model]. On-premise avoids cloud egress.

Submission package

DELIVERED

- Solution presentation: this deck, as PPTX and PDF
- Technical proposal and HLD: the eight required elements and the ten dimensions, with measured, modelled and estimated figures labelled, as PDF
- Workflow and integration diagram (SVG, PNG and PDF)
- Demo video 1, own feed: onboarding, detection, watchlist correlation, automatic alert
- Demo video 2, government-provided feed: viewing, ANPR with crops, vehicle and person detection, line crossing, detection report
- Registry API documentation (OpenAPI), sample onboarded dataset, gap-analysis report, detection report as CSV and HTML
- Source repository: Python 3.13 with FastAPI and SQLite for the demo, React with Leaflet and hls.js, YOLOX-S and PaddleOCR under Apache-2.0

LINKS

- Source repository: github.com/adityashroff06-code/sentinel-gujarat
- Demo video 1 (unlisted): to be added at submission
- Demo video 2 (unlisted): to be added at submission
- Hosted platform: to be added at submission; evaluator credentials go on the submission form only
- HLD (PDF): deliverables/HLD.pdf in the repository

Every link is opened from a private window before submission. The repository, its history, the videos and the screenshots are swept for credentials before submission.

Sentinel: one platform, five departments, watching is not storing.